

Automatic Depression Detection Among Higher Education Students Based on DeepFM

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Abstract—Depressive disorder has become a common problem among higher education students, but it often gets undiagnosed and untreated due to unrecognized symptoms, poor access to medical resources, and fear of stigma. To improve the situation, automatic depression detection would be essential. In this article, we explore the feasibility of depression detection in higher education students using their behavioral data automatically collected by the university system. First, a DeepFM network, which can not only take discrete–continuous mixed features as its input but also can learn linear and nonlinear relations between the input and the output, is presented for depression detection. A modified focal loss (MFL) function is then proposed to alleviate data imbalance impact caused by the fact that the proportion of healthy students outweighs those diagnosed with depression significantly. To verify the effectiveness of the proposed method, behavioral data from 3218 students were collected, of which 179 were diagnosed with depression by university psychologists using PHQ-9 scale scores. Fivefold cross validations are performed, and the experiment results have illustrated that DeepFM obtains the highest average accuracy compared with multilayer perceptron (MLP), factorization neural network (FNN), and product-based neural network (PNN), demonstrating the effectiveness of the proposed framework for depression detection among university students.

Index Terms—Campus big data, data imbalance, deep learning, DeepFM, depression detection.

NOMENCLATURE

X^s	Behavioral features of the s th student.
y^s	Ground truth of the s th student.
$\mathcal{X}_d^s$	Discrete features after one-hot coding of the s th student.
$\mathcal{X}_c^s$	Continuous features after min–max normalization of the s th student.

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$\mathcal{X}^s$	Input of the FM component and the DNN component.
F_{FM}	Output of FM component.
F_{DNN}	Output of DNN component.
p	Probability of depression of the s th sample.
$\hat{y}$	Detection result given by the model.
Θ	Parameters of the proposed model.
d	Dimension of $\mathcal{X}^s$.
v_i	i th row in V .
x_i	i th value of $\mathcal{X}^s$.
L	Modified focal loss function.
S_1, S_0	Number of students with depression, number of healthy students.
λ	Imbalance degree.
$\Delta \theta$	Gradient in backpropagation.
h_l	Number of neurons of the l th layer of DNN.
$a^{(l)}$	Output of the l th layer of DNN.
σ	Sigmoid function.
φ	ReLU function.

I. INTRODUCTION

DEPRESSION is a common illness worldwide, with an estimated 3.8% of the population affected, and it is a significant contributor to the overall global burden of disease [1]. Detection and diagnosis of depression early are crucial steps for proper treatment [2]. Recently, automatic objective assessment methods to assist mental health detection and diagnosis have been widely explored [3], [4], [5], [6].

Thus far, video analysis is regarded as the main approach for depression detection, since it highly correlates with head movements, facial expressions, gaze, and so on. For instance, Uddin et al. [7] proposed a two-stream deep spatiotemporal framework for depression-level prediction. They used a temporal median pooling approach and employed a multilayer bidirectional long short-term memory (Bi-LSTM)-based model for depression analysis on video data. Niu et al. [8] developed dual attention (DA) and element recalibration (ER) blocks to construct the dual attention and element recalibration (DAER) network, which was used to extract the facial representations of individuals with different depression levels. Yang et al. [9] proposed a multimodal framework for predicting the patient health questionnaire depression scale (PHQ-8 score) and classifying an individual as depressed or not depressed, by hybridizing deep models and shallow methodologies. de Melo et al. [10] proposed a deep learning architecture

called the maximization and differentiation network to represent facial expression variations for depression assessment. Although these aforementioned studies have demonstrated the feasibility of depression analysis using video data, such data may implicate serious privacy concerns. Besides video data, EEG and mobile phone data have also been explored recently for depression analysis. For instance, Acharya et al. [11] and Seal et al. [12] both proposed convolutional neural networks (CNNs) to automatically learn features to characterize depressed and normal EEG signals. Cai et al. [13] constructed a multimodal model to distinguish depressed patients from normal controls by fusing different EEG data sources, which were under neutral, negative, and positive audio stimulation. Shen et al. [14] presented an optimal channel selection method via kernel-target alignment (KTA) and its application in depression detection. Although the recent advancements in EEG make it a powerful tool, it is relatively intrusive and uncomfortable during EEG data collection. Similar work has also been reported to use intrusive phone metadata [15], [16], [17].

Social media have proven to be an unintrusive and stable source of data, and some researchers recently also demonstrated the possibility of depression detection using social media data. For instance, Choing et al. [18] propose 90 unique features, through a combination of feature extraction using sentiment lexicons and content-based features from the social media messages themselves, to detect depression using machine learning classifiers. Guo et al. [19] collected data from Sina Weibo and approached depression as a binary classification problem. The effectiveness of their approach was verified using classical machine learning methods. Angskun et al. [20] also explored the effectiveness of data from Tweets in detecting depression and found that machine learning models can capture depressive moods of depression sufferers. Tong et al. [21] proposed a classifier called cost-sensitive boosting pruning trees (CBPTs) and tested its performance on two publicly accessible Twitter depression detection datasets. Shen et al. [22] proposed a cross-domain deep neural network model with feature adaptive transformation and combination strategy (DNN-FATC) to enhance depression detection via social media with multisource datasets. Although social media can provide a nonintrusive channel for collecting depressed text data, many people may not use social media at all.

Unlike the data collection challenges faced by depression detection for the general public, the emergence of digitization in education has made various types of student data readily available [23]. In this article, we explore the feasibility of detecting depression in higher education students using the behavioral data automatically collected by the university system. With the behavioral data containing both discrete and continuous characteristics, the DeepFM model can automatically learn depression detection patterns among students. DeepFM is a deep learning structure for discrete–continuous mixed data as input, which has been widely used in multiple tasks, such as metal–organic properties prediction [24], passenger car sales prediction

prediction [25], taxi pickup area recommendation [26], and so on.

The main contributions of this study include the following.

- 1) A DeepFM network, which can not only take input for discrete–continuous mixed features but also learn both linear and nonlinear relations between the input and the output, is presented for depression detection.
- 2) A modified focal loss (MFL) function is then proposed to alleviate the data imbalance impact caused by the fact that the proportion of healthy students outweighs those diagnosed with depression significantly. The MFL can compensate for less training data for the depressed group by making it descend further in the stochastic gradient descent (SGD) process, thus reducing the bias of the model in the training phase.
- 3) To verify the effectiveness of the method, behavioral data from 3218 students were collected, of which 179 were diagnosed with depression. Fivefold cross validations were performed, and the experiment results illustrate that DeepFM has obtained the highest average accuracy 96.4% compared with multilayer perceptron (MLP) 93.3%, factorization neural network (FNN) 91.8%, and product-based neural network (PNN) 96.2%, which demonstrated the effectiveness of the proposed detection framework for enhancing depression detection performance.

The remaining article is arranged as follows. Section II introduces the materials and methods. In Section III, the experiment results are reported. Then, the discussion is presented in Section IV. The conclusion is presented Section V.

II. METHODOLOGY AND MATERIALS

In this section, we will introduce the proposed depression detection method: we will briefly describe the data preparation, and then, we explain the proposed depression detection framework, which includes a DeepFM network architecture and an MFL function. We will elaborate on each part below. For a better understanding of our proposed depression detection algorithm, Nomenclature summarizes the symbols/notations used in this article.

A. Data Preparation and Notations

Participants in the study were full-time undergraduates (aged from 19 to 21 and enrolled between 2015 and 2017) from our university. First, we distributed 12 000 questionnaires to all the students in these three cohorts, and 5000 students indicated they would like to participate in our study. After signing a nondisclosure agreement for data sharing with these student volunteers from the university, their behavioral data were downloaded from the University Data Center. Volunteers with more than 20% missing data and outliers (judged by Z-score [27]) were removed directly. For the remaining records, miceforest [28], a way of data interpolation, is performed to fill in missing values. Finally, the behavioral dataset containing 3218 undergraduates (2491 males and 727 females)

TABLE I
 DETAILS OF THE BEHAVIORAL DATA

Discrete Features (8)	Demographic Data (8)	Gender, Age, Nationality, Native place, Family number, Family financial situation, Year of entry, Major
Continuous Features (27)	Academic Performance Data (16)	Sum of Courses scores, Mean of Courses scores, Std of courses scores, GPA, Number of scholarship awarded, Level of scholarship awarded, Class ranking, Number of courses absence, Number of courses late, Number of courses attendance, Number of courses leaving in advance, Number of retaking course, Number of resitting course, Sum of missing classes hours, Mean of missing classes hours, Std of missing classes hours
	Physical Quality Data (7)	PE scores, BMI, Total number weekly workouts, Average number of weekly workouts, Std of number of weekly workouts, Total hours of workouts, Average hours of workouts
	Daily Data (4)	Number of canteen consumption, Number of companions, Number of late getting back to dorms, Number of stay up late

was created, and 179 of them have been diagnosed with depression by university psychologists. They used PHQ-9 scale [29] scores to determine if the student was suffering from depression. Students with a score greater than 4 are identified as depressed. Many studies have demonstrated that there is connection between depression and academic performance [30], [31], [32], daily behavior [33], [34], [35], [36], [37], movement/exercises [38], [39], [40], and demography [41], [42], [43], [44]. Therefore, demographic data, academic performance data, physical quality data, and daily life data were selected in this study. All these data were automatically collected by the University Data Center due to the recent advancement of university digitalization. The details of the data are listed in Table I. For any student s ($s = 1, 2, \dots, 3218$), his (or her) data record in the depression detection data set was depicted as (X^s, y^s) , where X^s is a vector of dimension 35 that indicates the features of the behavioral data and y^s denotes the ground truth ($y^s = 1$ means s is diagnosed with depression, and $y^s = 0$ means s is not depressed during the test). As shown in Table I, X^s consists of discrete features and continuous features. Since not all discrete features can be compared in their values, one-hot encoding is adopted to process them. After that, discrete features are converted to a high-dimensional sparse feature vector $\mathcal{X}_d^s$. For continuous features, min-max normalization was performed to get a feature vector $\mathcal{X}_c^s$.

B. Depression Detection Framework Based on DeepFM

The main framework of the depression detection model based on DeepFM is depicted in Fig. 1. First, due to the

 TABLE II
 DNN STRUCTURE. THE BATCH SIZE IS SET TO 64 IN OUR EXPERIMENTS. d IS THE DIMENSION OF THE INPUT VECTOR $\mathcal{X}^s$. THE OUTPUT OF FC4 IS F_{DNN}

Layer	Inputsize	Outputsize	Activation	Dropout
FC1	batchsize $\times d$	batchsize $\times 256$	ReLU	0.1
FC2	batchsize $\times 256$	batchsize $\times 128$	ReLU	0.1
FC3	batchsize $\times 128$	batchsize $\times 64$	ReLU	0.1
FC4	batchsize $\times 64$	batchsize $\times 1$	-	-

high-dimensional and extremely sparse characteristic of discrete feature $\mathcal{X}_d^s$, an embedding layer is used to reduce the dimension of the vector. Then, the dimension-reduced $\mathcal{X}_d^s$ is concatenated to the continuous feature $\mathcal{X}_c^s$, obtaining the feature $\mathcal{X}^s$, as the input of the model. $\mathcal{X}^s$ is taken as the input of the factorization-machine (FM) component and deep neural network (DNN) component, where FM learns linear relations and part nonlinear relations and DNN mainly learns nonlinear relations between the input and the output. Next, the outputs of FM (F_{FM}) and DNN (F_{DNN}) are summed up, and the summation is used as an input of the activation function sigmoid.

C. Architecture of DeepFM

Considering the discrete-continuous mixed characteristic of our dataset, DeepFM is used as a classifier in this study. The DeepFM network consists of two components: the FM component and the DNN component.

First, the FM component parameters were estimated as $w = [w_0, w_1, \dots, w_d]$ and a matrix $V \in \mathbb{R}^{d \times k}$. A row v_i within V is a k -dimensional vector, which describes the i th feature. The output of FM is defined as follows:

$$F_{\text{FM}} = w \begin{pmatrix} 1 \\ \mathcal{X}^s \end{pmatrix} + \sum_{i=1}^{d-1} \sum_{j=i+1}^d v_i v_j^T x_i x_j \quad (1)$$

where $\mathcal{X}^s = [x_1, x_2, \dots, x_d]$. For each feature $x_i \in \mathcal{X}^s$, a scalar $w_i \in w$ is used to weigh its order-1 importance, and a latent vector is used to weigh its importance of interactions with other features.

Second, the DNN, a feedforward neural network, is used to learn the nonlinear relations between the input and the output. The input of the DNN component is $\mathcal{X}^s$ as well. As shown in Fig. 1, the DNN component is mainly composed of four fully connected layers, and each fully connected layer is followed by a ReLU layer and a dropout layer. The detailed structure of DNN is shown in Table II. After forward propagation, the output of the DNN component F_{DNN} is obtained.

Finally, after the action of the sigmoid activation function, the output of the model, the probability of depression, is obtained. The details are as follows:

$$p = \sigma(F_{\text{FM}} + F_{\text{DNN}}) \quad (2)$$

where σ is the sigmoid activation function, p is the estimated probability of depression, and the predicted label of a student is defined as follows:

$$\hat{y} = \begin{cases} 1, & \text{if } p > 0.5 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

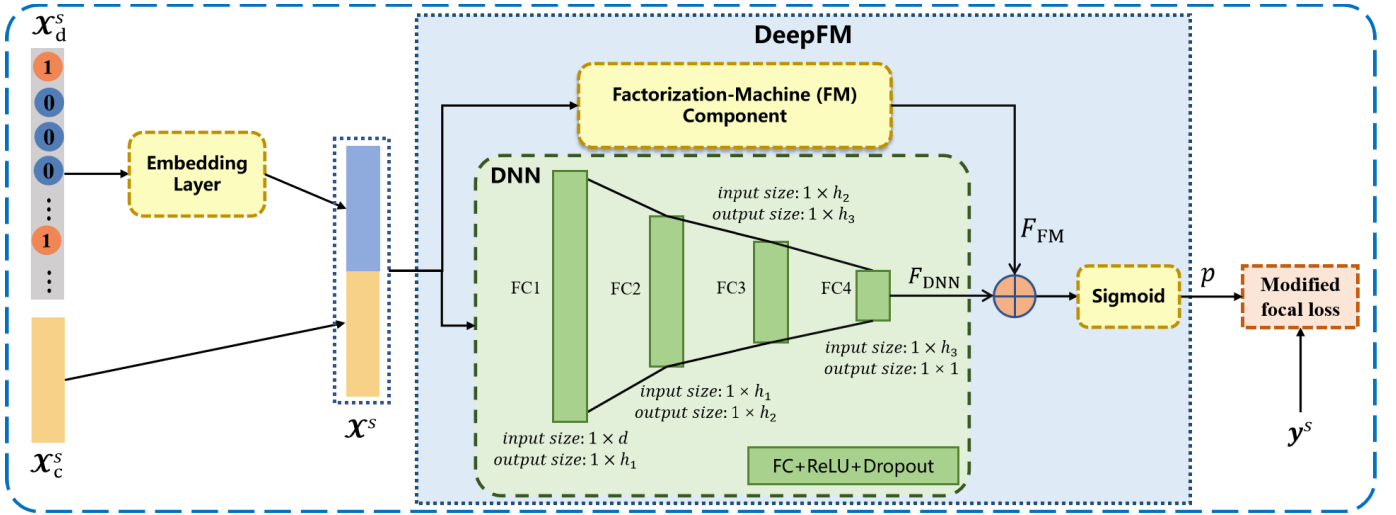


Fig. 1. Main framework of depression detection model. The framework mainly includes two components, FM and DNN, and the proposed loss function, the MFL function. The input of the framework is a vector $\mathcal{X}^s$ composed of depressive features. The output, p , is the probability of depression.

where $\hat{y}$ is the detection result given by the model ($\hat{y} = 1$ means the student is classified as depressed, and $\hat{y} = 0$ means the student is classified as healthy).

The model parameters $\Theta = \{\mathbf{w}, \mathbf{V}, \mathbf{U}\}$ are updated by SGD algorithm, where $\mathbf{w}$ and $\mathbf{V}$ are the parameters in the FM component and $\mathbf{U}$ represents the parameters in the DNN component.

D. Modified Focal Loss

To reduce the influence of the imbalance characteristics of the dataset, we proposed a class imbalance loss function named MFL to reduce the bias of the model. With the probability p obtained in (2) and the ground truth y^s defined in Section II-A, MFL is designed as follows:

$$L = -y^s \left(\left(1 - \sin\left(\frac{\pi}{2} \hat{p}\right) \right) \cos\left(\frac{\pi}{2} \hat{p}\right) f(\lambda, S_1, S_0) \right) (1 - \hat{p})^\gamma \times \log \hat{p} - (1 - y^s) \left(\left(1 - \sin\left(\frac{\pi}{2} \hat{p}\right) \right) \cos\left(\frac{\pi}{2} \hat{p}\right) \right) \times (1 - \hat{p})^\gamma \log \hat{p} \quad (4)$$

and

$$\hat{p} = \begin{cases} p, & \text{if } y^s = 1 \\ 1 - p, & \text{otherwise} \end{cases} \quad (5)$$

where $\sin()$ and $\cos()$, respectively, denote the sine function and cosine function, and γ is an adjustable parameter. The function $f()$ introduces the distribution of the training dataset into the loss function. In this way, the gradients of the minority class will descend further to make up for the disadvantage of less training data. $f(\lambda, S_1, S_0)$ is defined as follows:

$$f(\lambda, S_1, S_0) = \begin{cases} (S_0/S_1)^{\frac{1}{2}}, & \text{if } \lambda \leq 1 \\ (S_0/S_1)^{\frac{1}{8}}, & \text{otherwise} \end{cases} \quad (6)$$

where λ is the imbalance degree and is defined as follows:

$$\lambda = -\frac{1}{2} \log \frac{S_1}{S_0}. \quad (7)$$

Here, S_0 denotes the number of healthy students, and S_1 denotes the number of students with depression.

During the model training, any parameter $\theta \in \Theta$ in the model would be updated as follows:

$$\theta_{\text{new}} = \theta_{\text{old}} + \eta \times \Delta\theta \quad (8)$$

$$\Delta\theta = \frac{\partial L}{\partial \theta} \quad (9)$$

where θ_{old} denotes the value of parameter θ of the previous iteration and θ_{new} denotes the current value updated by SGD. η is the learning rate. $\Delta\theta$ denotes the current gradient. According to the chain rule

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial \hat{p}} \frac{\partial \hat{p}}{\partial p} \frac{\partial p}{\partial \theta} \quad (10)$$

$$\frac{\partial p}{\partial \theta} = p(1-p) \frac{\partial (F_{\text{FM}} + F_{\text{DNN}})}{\partial \theta} \quad (11)$$

$$= \begin{cases} 1, & \theta = w_0 \\ x_i, & \theta = w_i \\ x_i \sum_{e=1}^k v_{j,e} x_j - v_{i,e} x_i^2, & \theta = v_{i,e} \\ \left(\varphi' \left(\sum_{r=1}^{h_{l-1}} u_{r,g}^{(l-1)} a_r^{(l-1)} + b_r^{(l-1)} \right) a_r^{(l-1)} \right), & \theta = u_{r,g}^{(l-1)} \end{cases} \quad (12)$$

where φ is the ReLU function, $u_{r,g}^{(l-1)} \in \mathbf{U}^{(l-1)}$, $\mathbf{U}^{(l-1)} \in \mathbb{R}^{h_l \times h_{l-1}}$ is the weight matrix of the $(l-1)$ th layer of DNN component, h_l denotes the number of neurons of the l th layer, so $0 < r \leq h_{l-1}$ and $0 < g \leq h_l$, $a_r^{(l-1)} \in \mathbf{a}^{(l-1)}$ represents the output of the r th neuron of the $(l-1)$ th layer, and

$b_r^{(l-1)} \in \mathbf{b}^{(l-1)}$ represents the bias of the r th neuron of the $(l-1)$ th layer. Also, $a_r^{(l-1)} = \varphi(\sum_{z=1}^{h^{(l-2)}} u_{z,r}^{(l-2)} a_z^{(l-2)} + b_r^{(l-2)})$

$$\frac{\partial \hat{p}}{\partial p} = \begin{cases} 1, & \text{if } y^s = 1 \\ -1, & \text{otherwise} \end{cases} \quad (13)$$

$$\frac{\partial L}{\partial \hat{p}} = \begin{cases} \begin{pmatrix} -f(\lambda, S_1, S_0)(1 - \hat{p})^{\gamma-1} \\ ((1 - \hat{p}) \log \hat{p} \\ \times (\sin(\frac{\pi}{2} \hat{p}) + \cos(\pi \hat{p})) \\ \times (-\frac{\pi}{2}) \\ + (\frac{(1 - \hat{p})}{\hat{p} \ln 10} - \gamma \log \hat{p}) \\ \times (1 - \sin(\frac{\pi}{2} \hat{p})) \cos(\frac{\pi}{2} \hat{p})) \end{pmatrix}, & \text{if } y^s = 1 \\ \begin{pmatrix} -(1 - \hat{p})^{\gamma-1} \\ \times ((1 - \hat{p}) \log \hat{p} \\ \times (\sin(\frac{\pi}{2} \hat{p}) + \cos(\pi \hat{p})) \\ \times (-\frac{\pi}{2}) \\ + (\frac{(1 - \hat{p})}{\hat{p} \ln 10} - \gamma \log \hat{p}) \\ \times (1 - \sin(\frac{\pi}{2} \hat{p})) \cos(\frac{\pi}{2} \hat{p})) \end{pmatrix}, & \text{otherwise.} \end{cases} \quad (14)$$

From (8) to (14), we can conclude that the length of gradient of positive class ($y^s = 1$) is $f(\lambda, S_1, S_0)$ ($f(\lambda, S_1, S_0) > 1$) times as long as the negative class ($y^s = 0$), which indicates that the minority class will descend further during model training. Besides, the value of function $f()$ is not as large as possible. This is because an enormous gradient will result in gradient vanishing, loss not being declined, and unstable output.

E. Hyperparameter Setting of Our Proposed Framework

As shown in Table II, the sizes of the outputs from FC1 to FC3 are 256, 128, and 64, respectively. The output size of the last layer of DNN is 1. In the training phase, the batch size was set to 64, and the learning rate was set to 0.01. We used SGD for training, and the dropout rate in each dropout layer was 0.1. The length of v_i and k (proposed in Section II-C) was set to 8. In addition, the parameter γ in the MFL was set as 3. Fivefold cross validations with each fold of ten random runs (using different random seeds) were performed. Therefore, in each experiment, we use 80% of the data (i.e., 2574 students, diagnosed versus undiagnosed: 143 versus 2431) for training and 20% (i.e., 644 students, diagnosed versus undiagnosed: 36 versus 608) for testing.

F. Evaluation Metrics

In this study, in addition to classification accuracy, sensitivity, specificity, G-mean, F1 score, and area under receiver operating characteristic curve (AUC) were calculated to evaluate the performance of the proposed method. Sensitivity and specificity represent the proportion of all positive samples predicted correctly to all actual positive samples and the

proportion of all negative samples predicted correctly to all actual negative samples, respectively. In this article, positive samples are diagnosed students, and negative samples are healthy students. F1 score, G-mean, and AUC are the balance of two evaluation indexes, which are commonly used in imbalance classification studies [45]. They are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (15)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (16)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (17)$$

$$\text{G-mean} = \sqrt{\text{Sensitivity} \times \text{specificity}} \quad (18)$$

$$\text{F1 score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{Precision} + \text{recall}} \quad (19)$$

where precision = $\text{TP}/(\text{TP} + \text{FP})$ and recall = $\text{TP}/(\text{TP} + \text{FN})$. The terms TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

III. EXPERIMENTAL RESULTS

In this section, we first introduce the baseline methods. Then, we evaluate the performance of the proposed method on the student behavioral dataset. Extensive comparisons of classification performance are implemented between the proposed method and other neural networks. Finally, the influence of the proposed loss function, such as the convergence of the loss function, the influence of the parameters in the modified loss function, and comparisons with other loss functions, was also reported.

A. Baseline Methods

To verify the advantages of DeepFM in processing discrete–continuous mixed features, we compared the performances with FNN [46], PNN [47], and MLP [48]. The structure of MLP is the same as that of the DNN component. So, the performance of MLP expresses the role of high-order feature interactions. FNN is a deep learning model for predicting user behavior. To minimize the influence of other factors, the structure of the fully connected layers in FNN is set to be the same as that of MLP. PNN was a deep learning model based on multiplication to represent feature crossing. The PNN network structure adds the product layer to the traditional DNN, so as to realize the crossover of features. Similarly, the fully connected network portion of the PNN in this document was structured in the same way as the MLP setup. In the training phase, all of the models updated their parameters according to the loss given by the MFL function.

B. Depression Detection Performance

The depression detection performance comparison was conducted between the proposed method DeepFM with the MFL (DeepFM-MFL), and MLP-MFL, FNN-MFL, and PNN-MFL. All of them were implemented with PyTorch¹ on our

¹<https://pytorch.org/>

TABLE III

COMPARISON OF CLASSIFICATION RESULTS FOR DIFFERENT MODELS (MEAN $\pm$ STD UNIT: %). “A-B” REPRESENTED MODEL A WITH LOSS FUNCTION B DURING THE TRAINING PHASE. “MFL”: MODIFIED FOCAL LOSS FUNCTION

Models	Sensitivity	Specificity	F1score	AUC	GM	Accuracy
FNN-MFL	65.4 $\pm$ 15.8	94.4 $\pm$ 1.3	49.1 $\pm$ 7.9	84.1 $\pm$ 8.6	59.8 $\pm$ 8.4	91.8 $\pm$ 2.7
MLP-MFL	68.6 $\pm$ 10.3	95.6 $\pm$ 0.9	52.6 $\pm$ 16.2	86.4 $\pm$ 5.3	61.8 $\pm$ 8.4	93.3 $\pm$ 1.3
PNN-MFL	69.4 $\pm$ 10.3	97.3 $\pm$ 1.8	74.3 $\pm$ 8.7	84.3 $\pm$ 7.4	80.2 $\pm$ 7.3	96.2 $\pm$ 2.4
DeepFM-MFL	82.1$\pm$7.7	99.0$\pm$0.1	81.2$\pm$5.6	91.9$\pm$4.5	86.9$\pm$8.2	96.4$\pm$0.8

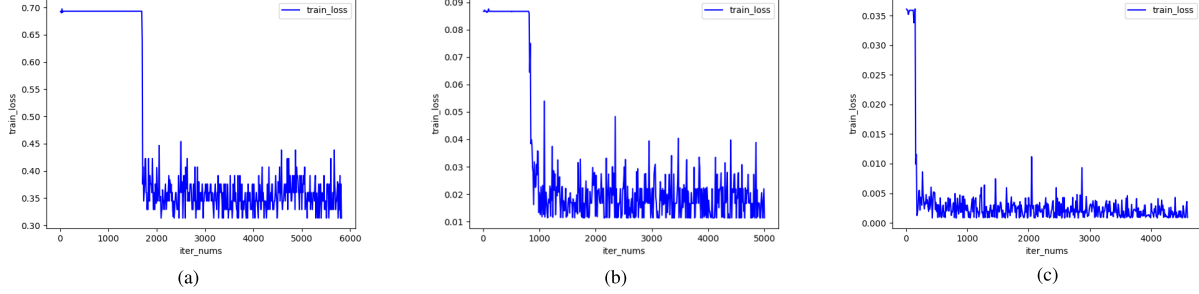


Fig. 2. Different loss function curves during the DeepFM model training. The horizontal axis represents the number of iterations “iter_nums,” and the vertical axis represents the current loss value. (a) CE loss. (b) FL. (c) MFL (proposed).

Dell PC with the Intel² Core³ i7-10700 CPU @ 2.90 GHz, 16-GB RAM, and 64-bit Windows 10 OS using Pycharm 2020.1.2 $\times$ 64. The average classification accuracy, sensitivity, F1 score, AUC, G-mean, and specificity were calculated over fivefold cross validation with ten runs (each run with a different random seed) to evaluate the detection performance of these methods. Table III illustrates the classification results across all students in the test set. As we can see from the table, the proposed DeepFM-MFL method can achieve the best detection performance compared with the others with a sensitivity of 82.1% $\pm$ 7.7%, an F1 score of 81.2% $\pm$ 5.6%, an AUC of 91.9% $\pm$ 4.5%, a GM of 92.3% $\pm$ 7.9%, an accuracy of 96.4% $\pm$ 0.8%, and a specificity of 99.0% $\pm$ 0.1%.

C. Influence of the MFL

In order to further evaluate the performance of the proposed MFL, in this part, first, we study the convergence of the loss functions, the role of $f()$ function, and the influence of the γ parameter in the loss function. Then, the classification performances of cross-entropy loss (CE loss), focal loss (FL) [49], and the proposed loss (MFL) were compared. In this article, the parameter γ of $(1 - \hat{p})^\gamma$ in FL was set to 3 (equals to γ of MFL).

1) *Convergence of the Proposed Framework:* To demonstrate the convergence of the proposed framework, we illustrate the convergence process of the MFL during the DeepFM model training phase in Fig. 2(c). As shown in Fig. 2(c), the proposed loss begins to converge with large fluctuations after 200 iterations. After 3000 iterations, the fluctuation decreases, and the convergence becomes stable. By contrast, Fig. 2(a) shows the convergence process of CE loss under the same conditions as the MFL. As shown in Fig. 2(a), it is not until 1600 iterations that the CE loss begins to converge, and the loss curve fluctuates greatly, which indicates that the output of the model is very unstable. Also, Fig. 2(b) shows the FL curve. In short, comparing the three figures in Fig. 2, the CE

loss curve and FL curve have more vibration than the proposed loss curve. Besides, the proposed loss function converges faster than others. We conjecture that the main reason is that our loss function takes into account the distribution of the training data, so that the minority class will get more attention during the model training.

2) *Influence of the Parameters in MFL:* The main idea of the MFL is that it introduces the distribution of the training dataset into the loss function by defining the function $f()$. Therefore, in order to investigate the role of the function $f()$, we, respectively, set f as 1 and $f()$ [whose value is calculated by (6)]. The results of four models (mentioned in Section III-A) are shown in Fig. 3. This figure shows that sensitivity and G-mean are improved when $f = f()$. Next, we analyze the impact of the value of γ in (4). From (4), we can see that with the increase of the value of γ , the value of loss will decrease. In this way, well-classified samples will get a smaller loss, so that the model will pay less attention to these samples but more attention to others during the model training. Fig. 4 shows how the values of sensitivity and G-mean change with the value of γ . As can be seen from the figure, both sensitivity and G-mean reach their maximum when γ is equal to 3.

3) *Classification Performance Comparisons of Different Loss Functions:* In this part, we compare the classification performances of CE loss, FL, and the proposed loss on MLP, FNN, PNN, and DeepFM models. Fig. 5 shows the sensitivity values of four models with different loss functions. The figure demonstrates that the proposed loss function is superior to CE loss and FL under the depression detection task, which significantly improves the models’ ability to identify positive samples (which refer to students who have been diagnosed with depression in this article). Fig. 6 gives information about the distribution of estimated probabilities of depression of samples in a test set in a randomized experiment on the DeepFM model with three different loss functions. We can see immediately from the figure that more points are distributed above 0.5 when MFL is used in classification models, which indicates that more positive samples are found. That is the reason why the sensitivity is improved.

²Registered trademark.

³Trademarked.

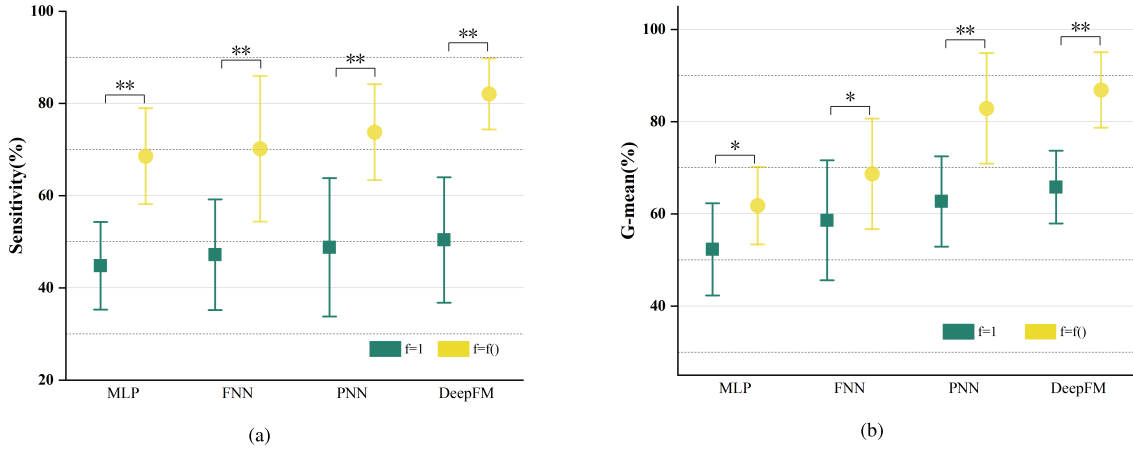


Fig. 3. Performance comparisons under different (a) sensitivity results and (b) G-mean results values of f . When $f = 1$, according to (4), the MFL would ignore the distribution of training data. When $f = f()$, according to (6), the MFL would consider the distribution of training data. The asterisks indicate significant differences between the seven methods obtained by paired t-tests (* $p < 0.05$ and ** $p < 0.01$).

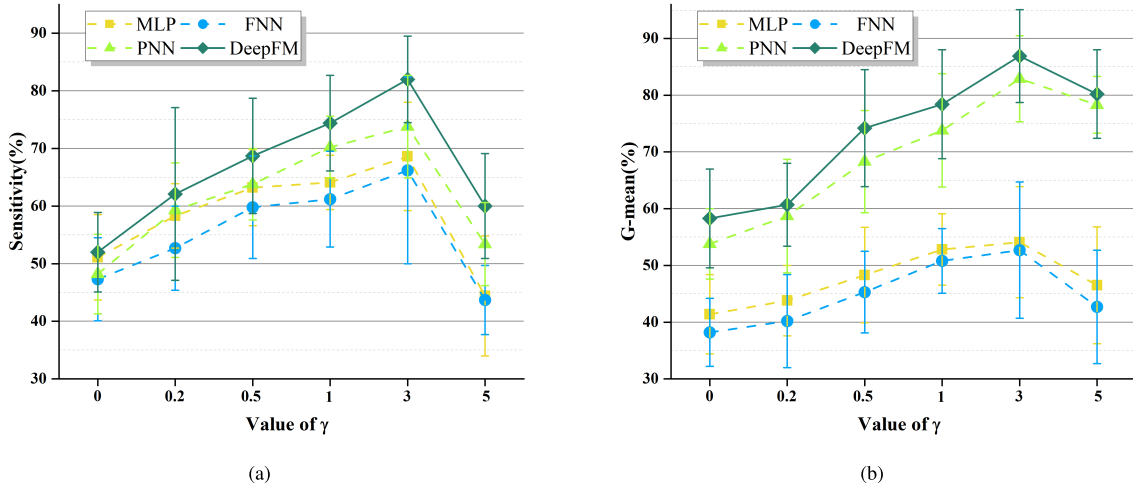


Fig. 4. Performance comparisons under different (a) sensitivity results and (b) G-mean results values of the parameter γ in (4).

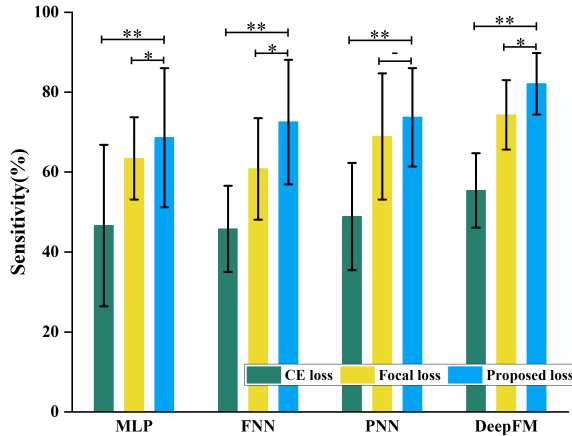


Fig. 5. Sensitivity comparisons of different loss functions. The asterisks indicate significant differences between the seven methods obtained by paired t-tests (* $p < 0.05$ and ** $p < 0.01$).

IV. DISCUSSION

A. Effect of Different Types of Data on Depression Detection

We further investigated the importance of four types of data (as shown in Table I) in determining student depression via an ablation study. We trained the model using three of the four types of data mentioned in Table I to investigate

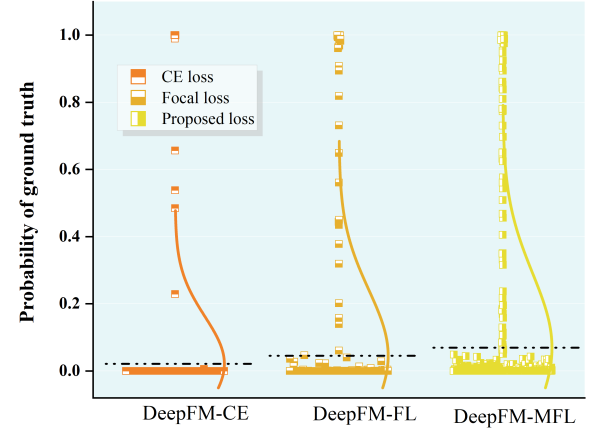


Fig. 6. Scatter diagram and normal distribution curve of estimated probability of ground truth of test set for DeepFM model with different loss functions. “A-B”: A denotes the chosen classification model, and B denotes the chosen loss function. “CE”: cross-entropy loss function. “FL”: focal loss function. “MFL”: modified focal loss function.

the importance of the remaining type of data. In this way, the larger the decrease in detection performance compared with the model trained using all types of data, the more important that type of data are in determining whether the student is depressed. Fig. 7 illustrates the averaged sensitivity across five-fold cross validation with ten runs (each run with a different

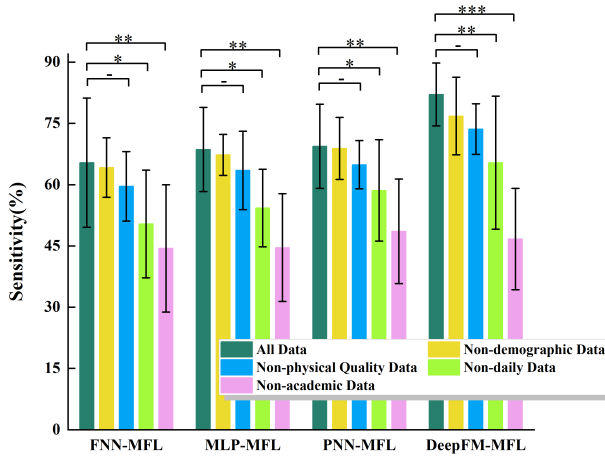


Fig. 7. Performance comparisons under different data. “All data” mean all of the data shown in Table I were used during model training. “Nontype data” mean apart from this type of data, the remaining data were used for model training. The asterisks indicate significant differences between the five conditions obtained by paired t-tests (* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$).

random seed). As can be seen from the figure, all four types of data we collected are crucial for the detection of depression among students, because removing any one of them would negatively affect the results of the experiment (sensitivities are reduced). The four types of data in descending order of importance are academic data, daily data, physical quality data, and demographic data. The results of the significance analysis show that academic data and daily data have a more significant effect on depression detection. This illustrates that depression among higher education students is mainly manifested through academic performance and some daily behaviors.

B. FM Versus DNN Versus DeepFM

To investigate the role of FM and DNN, we trained them separately and tested their performance. As we can see from Fig. 8, DeepFM has achieved the highest sensitivity compared with FM and DNN. In addition, FM outperforms DNN, indicating that the linear correlation between behavioral data and students’ depression is more pronounced compared with nonlinearity. Besides, in order to estimate the complexity of FM, DNN, and DeepFM, we compared their training time and test time. The average training times for FM, DNN, and DeepFM are 11.851, 28.048, and 54.486 s, respectively. Their average test times are 0.034, 0.067, and 0.062 ms, respectively. All of these experiments were performed on a Dell PC with the Intel Core i7-10700 CPU @ 2.90 GHz, 16-GB RAM, and 64-bit Windows 10 OS using Pycharm 2020.1.2 ×64. We would like to point out that although DeepFM has a more complicated training process than FM and DNN, and the training time is a bit longer, it will not affect the computational speed of the target detection during the test phase.

C. Comparisons With Similar Studies

To the best of our knowledge, we are the first to collect data for student depression detection in a campus big data scenario. Also, there are no similar studies using university data for depression detection, so our dataset type is unique at this time. In this case, it is not fair to compare our approach

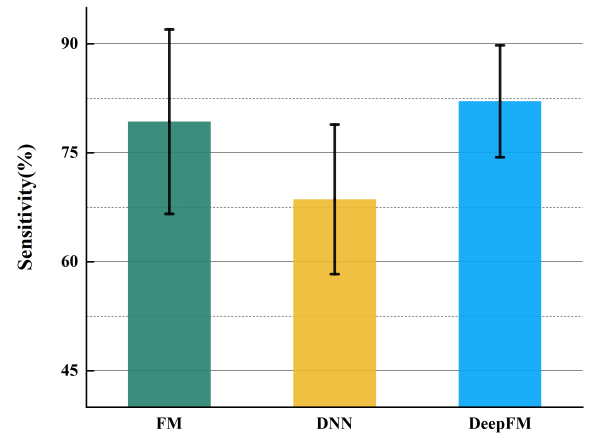


Fig. 8. Sensitivity comparisons of the FM component, the DNN component, and DeepFM.

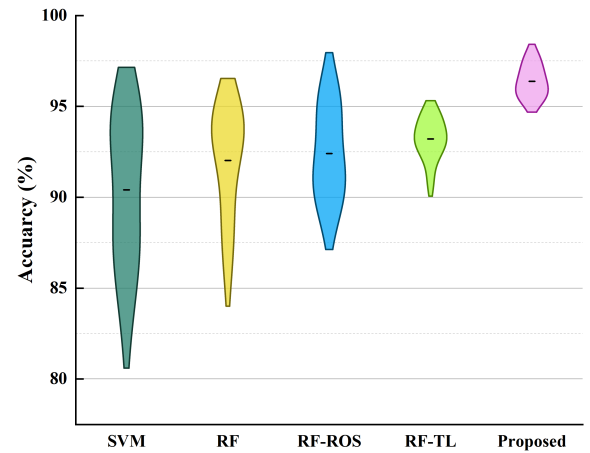


Fig. 9. Violin plots represent the distributions of classification accuracy achieved by five methods. The thick black line in each violin indicates the mean value.

with other studies, because we use different data. Although that said, there are several methods, i.e., random forest (RF), support vector machine (SVM) [19], [20], random forest with random oversampling (RF-ROS), random forest with Tomek link (RF-TL), and random forest with random oversampling and Tomek link (RF-ROSTL) [17] that might apply to our data. We, thus, implemented them and tested them on our dataset. Fig. 9 illustrates the averaged classification accuracy across fivefold cross validation with ten runs (each run with a different random seed). As we can see from the figure, the proposed DeepFM-MFL method can, in general, achieve the highest accuracy with regards to others.

D. Comparisons Among Diverse Student Populations

To explore whether there exists a significant difference between diverse student populations, we conducted some ablation studies on the feature of “gender” and “year of entry,” respectively. The distributions of “gender” and “year of entry” are shown in Table IV. We find that the depression rate of females is higher than that of males. The male base is larger than the female base, which may be due to the fact that the ratio of males to females in this university is 7:3. There is no significant difference in the number of students from different “year of entry.” Figs. 10 and 11 show the depression detection performance between different “gender” and “year of entry” on

TABLE IV
DISTRIBUTIONS OF GENDER AND YEAR OF ENTRY

Lable	Gender		Year of entry		
	Male	Female	2015	2016	2017
Depressed	120	59	56	63	60
Normal	2371	668	1030	996	1013

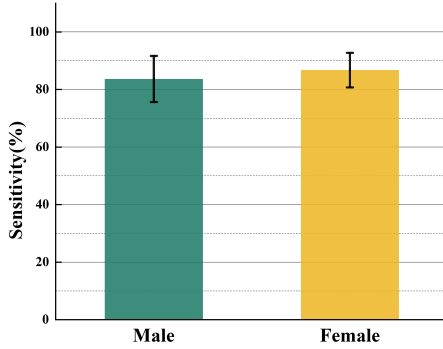


Fig. 10. Sensitivity comparisons between different “gender” on DeepFM-MFL.

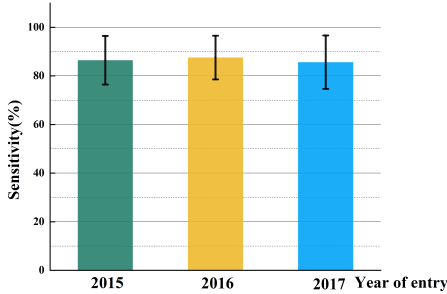


Fig. 11. Sensitivity comparisons between different “year of entry” on DeepFM-MFL.

the proposed model, which can be seen from the two figures that there indeed do not exist significant differences across “gender” and “year of entry.”

E. Future Work

In this study, we propose a new method to detect depression among higher education students. Real-world student behavioral data are used to detect student depression. Also, we have achieved a high average accuracy of 96.4%. However, there is still room for improvement, and optimization can be done in both method and application.

First, for the task of imbalanced classification, apart from the class imbalance loss function proposed in this article, an algorithm-level method, the imbalance classification problem can also be addressed at the data level. For example, generative models [50], [51], [52] can synthesize samples from the minority class. Besides, oversampling methods [53], [54], [55], [56] can also be used to achieve data augmentation. In the future, data augmentation can be used at the data level to further address the imbalance problem. Second, identifying the levels of students’ depression is definitely important for depression detection and is beneficial. In situations of scarce healthcare resources, major depressive disorder deserves primary attention as it is a significant contributor to poor academic performance, substance abuse, and attempted and completed suicide [57], [58], [59]. However, we have not achieved this in this preliminary study drawback as its main

purpose is to validate the feasibility of depression detection using the data automatically collected by the university system. Finally, in the next phase, we will also further process the data to extract the features that indicate students’ behavioral changes, verify the role of students’ behavioral changes in depression detection, and design a model to capture the characteristics of such changes.

V. CONCLUSION

This study presented a preliminary study of depression detection using student behavioral data automatically collected by the university system. DeepFM-MFL was proposed to enhance depression detection performance via learning linear and nonlinear relations between student behavioral data and depression, and MFL was proposed to alleviate the data imbalance caused by the fact that the proportion of healthy students outweighed those diagnosed with depression significantly. The experiment results have illustrated that the proposed DeepFM-MFL obtained the best performance compared with MLP, FNN, and PNN.

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